

Voice Activity Detection with Handcrafted Features and LBP Encoding

Nguyen Van Vinh¹, Nguyen Trung Dung², and Vu Van Quoc Huy³

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Abstract. Voice Activity Detection (VAD) is an important preprocessing step in speech processing systems such as automatic speech recognition and voice assistants. The goal of VAD is to distinguish speech from non-speech signals including silence, noise, and music.

In this work, we propose a machine learning-based VAD pipeline using handcrafted acoustic features. MFCC, LFCC, and STFT features are extracted from audio signals and further encoded using Local Binary Pattern (LBP) to capture local spectral patterns. The resulting features are classified using ensemble learning models including Random Forest and XGBoost.

Experiments are conducted using the MUSAN dataset for training and the Aurora 2 dataset for evaluation under noisy conditions. The results show that Random Forest achieves an accuracy of 0.993 on MUSAN, while XGBoost reaches 0.997. On the Aurora 2 dataset, the models achieve accuracies of 0.90 and 0.93 respectively. These results demonstrate that the proposed approach provides reliable performance for voice activity detection.

Keywords: Voice Activity Detection, Handcrafted Features, MFCC, LFCC, STFT, LBP, Machine Learning

1 Introduction

Voice Activity Detection (VAD) is a fundamental component in many speech processing systems such as automatic speech recognition, speaker verification, and voice assistants. The objective of VAD is to identify speech segments while removing non-speech signals such as silence, background noise, and music.

In this work, we propose a machine learning-based pipeline for voice activity detection. The system first performs preprocessing on audio signals and then extracts acoustic features from short frames. Three commonly used feature representations are investigated: Mel-Frequency Cepstral Coefficients (MFCC), Linear Frequency Cepstral Coefficients (LFCC), and Short-Time Fourier Transform (STFT).

To capture additional local spectral information, Local Binary Pattern (LBP) encoding is applied to the extracted features. The resulting features are then used to train ensemble learning models for speech and non-speech classification. In particular, Random Forest and XGBoost are employed due to their strong performance in classification tasks.

Experiments are conducted using the MUSAN dataset for training and the Aurora 2 dataset for evaluation in noisy environments. The results demonstrate that the proposed method achieves strong performance across different feature representations.

2 Related Work

VAD has been widely studied as a core module in speech-processing systems. Early approaches were primarily based on signal-level descriptors such as frame energy, zero-crossing statistics, and spectral measurements [5]. While computationally lightweight, these methods often degrade in noisy environments or in scenarios where non-speech signals are acoustically similar to speech.

Machine-learning approaches mitigate these limitations by learning decision boundaries directly from data. Prior studies have shown that MFCC features combined with SVM can effectively separate speech and non-speech classes [1]. Ensemble SVM variants have also been investigated to improve robustness through model diversity [2].

Research on noise-robust VAD further emphasizes the importance of pre-processing and feature design [3]. More recently, compact deep neural architectures have demonstrated competitive performance while maintaining manageable model complexity [4].

Motivated by these findings, this paper adopts a handcrafted-feature and ensemble-learning framework for VAD.

3 Methodology

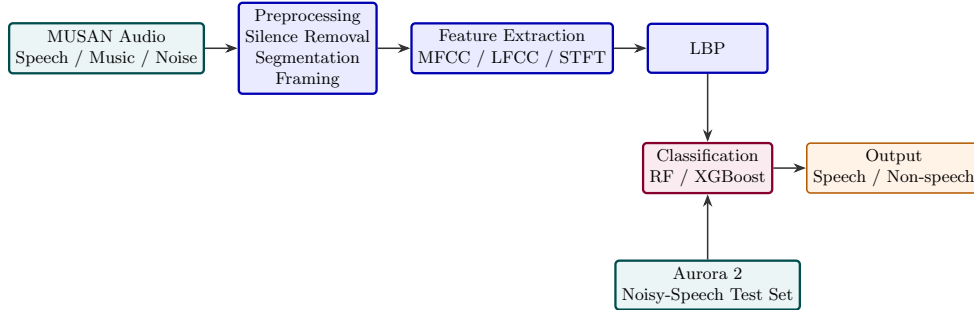


Fig. 1. Overview of the proposed Voice Activity Detection pipeline.

As illustrated in Fig. 1, the proposed VAD system consists of three main stages: preprocessing, handcrafted feature extraction, and machine learning-based classification. Audio signals are first processed to remove silence and normalize the waveform. Acoustic features are then extracted and used to train classification models that distinguish speech from non-speech segments.

3.1 Dataset

Two datasets are used in the experiments to train and evaluate the proposed VAD system.

MUSAN: The MUSAN dataset is a widely used audio corpus containing speech, music, and noise recordings. The dataset includes approximately 60 hours of speech, 42 hours of music, and 7 hours of noise. In this study, the music and noise subsets are grouped together and treated as the non-speech class, while the speech subset represents the speech class. MUSAN is used as the primary dataset for training and testing the classification models.

Aurora 2: Aurora 2 is a noisy speech dataset designed to evaluate the robustness of speech processing systems under adverse acoustic conditions. The dataset is created by adding various types of noise to the TIDigits corpus. It contains 8,440 training utterances and approximately 4,000 testing utterances. In this work, Aurora 2 is used to evaluate the generalization capability of the proposed VAD system in noisy environments.

3.2 Feature Extraction

To represent the acoustic characteristics of speech signals, three handcrafted feature types are extracted from short audio frames: MFCC, LFCC, and STFT. Additionally, LBP is applied to encode local structural patterns in the extracted features.

Mel-Frequency Cepstral Coefficients (MFCC): is one of the most widely used feature representations in speech and audio processing. It models human auditory perception by mapping the linear frequency spectrum onto the Mel scale. The MFCC extraction process typically involves computing the Short-Time Fourier Transform (STFT), applying a Mel-scaled filter bank, taking the logarithm of filter bank energies, and performing a Discrete Cosine Transform (DCT) to obtain compact cepstral coefficients that describe the spectral envelope of speech [8].

Linear Frequency Cepstral Coefficients (LFCC): is a cepstral feature similar to MFCC but uses linearly spaced filter banks instead of Mel-scaled filters. By preserving the linear frequency distribution, LFCC can retain more detailed information in higher frequency regions of the spectrum. This property makes LFCC useful in applications such as speaker verification and spoofing detection, where high-frequency artifacts may provide discriminative information [9].

Short-Time Fourier Transform (STFT) a time-frequency analysis technique used to represent non-stationary signals such as speech. The audio waveform is divided into short overlapping frames, and each frame is transformed into the frequency domain using the Fourier transform. The resulting spectrogram describes how spectral content changes over time and provides important information about the temporal and spectral structure of speech signals [7].

Local Binary Pattern (LBP): is a texture descriptor used to capture local spatial patterns in images. It works by comparing the intensity of a central pixel with its neighboring pixels and encoding the results as a binary number that represents the local texture. By computing the histogram of these binary patterns over an image or feature map, LBP effectively describes structural variations and has been widely used in pattern recognition tasks such as texture classification, face recognition, and analysis of time–frequency representations like spectrograms or MFCCs.

The input audio waveform is first transformed into different spectral representations using Mel-Frequency Cepstral Coefficients, Linear Frequency Cepstral Coefficients, and Short-Time Fourier Transform. STFT provides a time–frequency representation of the signal, while MFCC and LFCC represent the cepstral characteristics derived from the spectral domain. Each audio segment is converted into a feature matrix with shape (number of frame, number of coef Fig. 2). Next, Local Binary Pattern is applied to these feature matrices to capture local structural patterns. The resulting binary patterns are then summarized by computing a histogram, producing a 256-dimensional LBP (Fig. 3) feature vector that serves as the final representation for classification.

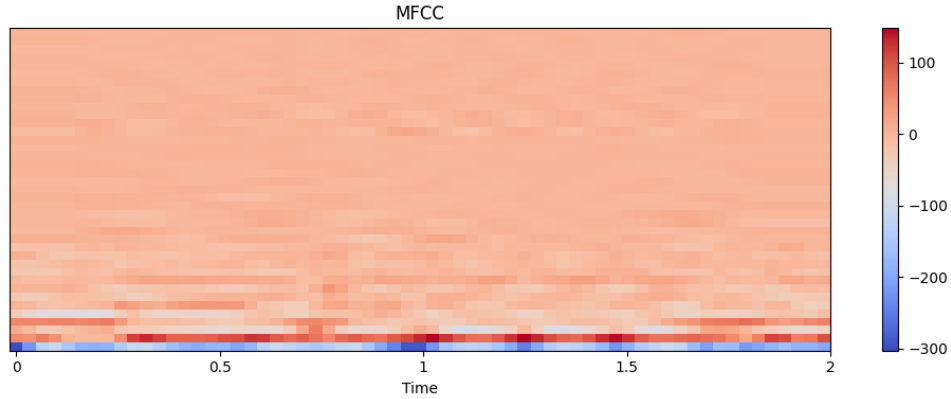


Fig. 2. Output of sample after extract with handcrafted feature (MFCC, LFCC, STFT)

3.3 Classification Models

To classify the extracted acoustic features, two ensemble learning models are employed: Random Forest (RF) and Extreme Gradient Boosting (XGBoost). These models are widely used in machine learning tasks due to their strong generalization capability and robustness when handling high-dimensional features.

Random Forest (RF) Random Forest is an ensemble learning algorithm based on the bagging technique, where multiple decision trees are constructed during

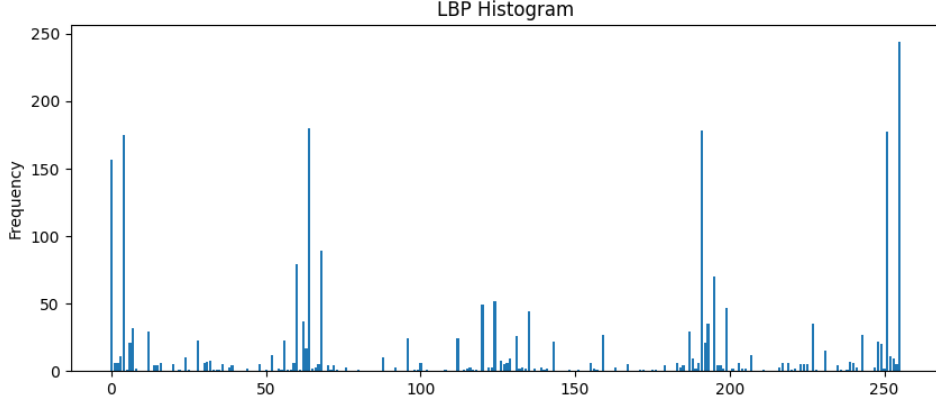


Fig. 3. Output after using LBP

training and the final prediction is obtained through majority voting. Each tree is trained using a randomly sampled subset of the training data and a random subset of features, which helps reduce overfitting and improves generalization performance [10].

Extreme Gradient Boosting (XGBoost) XGBoost is an optimized implementation of gradient boosting that builds an ensemble of decision trees sequentially. In this approach, each new tree is trained to correct the errors made by the previous trees, resulting in improved predictive performance. XGBoost incorporates several advanced techniques such as regularization, parallel processing, and efficient handling of sparse data, which enhance both training speed and model accuracy [11].

4 Experiments Setup

4.1 Preprocessing

The experimental pipeline consists of four stages: audio preprocessing, feature extraction, model training, and evaluation.

Data Preprocessing Two key preprocessing steps are reported:

– **Silence removal:**

- Each frame uses a 25 ms window with 15 ms overlap..
- The energy of each frame is computed as:

$$E = \sum_{n=0}^{N-1} x[n]^2$$

- Low-energy frames are then removed.

- **Audio Segmentation:** To increase the number of samples, the recordings were divided into fixed-length segments of 2 seconds. Each segment could contain speech or non-speech sounds. Subsequently, each segment is divided into overlapping frames with a 25 ms window and a 15 ms overlap for feature extraction (MFCC, LFCC, STFT).

After preprocessing, the dataset statistics reported in the source document are:

- Speech duration: 58.29 hours
- Non-speech duration: 46.35 hours

4.2 Evaluation:

The reported protocol uses 5-fold cross-validation. Performance is assessed by Accuracy, Precision, Recall, F1-score, and EER.

5 Results

We report three evaluation scenarios:

1. Train/test on Musan dataset

Table 1. Results of RandomForest

Feature	ACC	Prec	Recall	F1	EER%
MFCC static	0.978	0.982	0.974	0.978	2.1
MFCC static + Δ + $\Delta\Delta$	0.971	0.975	0.967	0.971	2.7
LFCC static	0.974	0.976	0.973	0.974	2.5
LFCC static + Δ + $\Delta\Delta$	0.976	0.977	0.976	0.976	2.3
STFT	0.993	0.994	0.993	0.993	0.63

Table 2. Results of XGBoost

Feature	ACC	Prec	Recall	F1	EER%
MFCC static	0.985	0.986	0.984	0.985	1.4
MFCC static + Δ + $\Delta\Delta$	0.982	0.984	0.981	0.982	1.7
LFCC static	0.980	0.980	0.981	0.981	1.9
LFCC static + Δ + $\Delta\Delta$	0.981	0.982	0.981	0.982	1.8
STFT	0.997	0.997	0.997	0.997	0.27

2. Train on Musan dataset and test on Aurora 2 Dataset

Table 3. XGBoost Results

Feature	ACC	Prec	Recall	F1	EER%
MFCC static	0.54	1	0.54	0.70	null
MFCC static + Δ + $\Delta\Delta$	0.62	1	0.62	0.76	null
LFCC static	0.73	1	0.73	0.84	null
LFCC static + Δ + $\Delta\Delta$	0.67	1	0.67	0.80	null
STFT	0.90	1	0.90	0.94	null

Table 4. RandomForest Results

Feature	ACC	Prec	Recall	F1	EER%
MFCC static	0.60	1	0.60	0.75	null
MFCC static + Δ + $\Delta\Delta$	0.75	1	0.75	0.86	null
LFCC static	0.74	1	0.74	0.85	null
LFCC static + Δ + $\Delta\Delta$	0.78	1	0.78	0.88	null
STFT	0.93	1	0.93	0.963	null

6 Discussion

The experimental results highlight several observations regarding the effectiveness of different handcrafted features for voice activity detection.

In the in-domain evaluation on the Musan dataset, all feature types achieved high performance, with accuracy above 0.97 for most configurations. However, STFT consistently provided the best results. In particular, XGBoost with STFT achieved an accuracy of 0.997 and an EER of 0.27%. This indicates that the full spectral representation of STFT preserves more discriminative information compared to cepstral features such as MFCC and LFCC.

Interestingly, adding dynamic features (Δ and $\Delta\Delta$) did not significantly improve performance and in some cases slightly reduced accuracy. This suggests that static spectral information may already be sufficient for frame-level speech/non-speech classification when using tree-based models.

In the cross-dataset experiment (training on Musan and testing on Aurora 2), the performance of MFCC and LFCC dropped considerably, indicating sensitivity to dataset mismatch. In contrast, STFT maintained much higher performance, achieving 0.90 accuracy with XGBoost and 0.93 with RandomForest. This suggests that STFT features provide better generalization across different acoustic conditions.

Overall, the results demonstrate that STFT features combined with tree-based classifiers offer a strong and robust baseline for voice activity detection.

7 Conclusion

This paper presents a machine learning-based approach for voice activity detection using handcrafted acoustic features and ensemble learning models. The proposed pipeline combines preprocessing, spectral feature extraction, LBP encoding, and supervised classification.

Experiments on the MUSAN and Aurora 2 datasets show that the proposed method achieves strong performance. Random Forest achieves an accuracy of

0.993 on MUSAN, while XGBoost reaches 0.997. On the Aurora 2 dataset, the models achieve accuracies of 0.90 and 0.93 respectively.

The results indicate that combining spectral features, LBP encoding, and ensemble learning models provides an effective solution for voice activity detection, especially in noisy environments.

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